

# JAY AGRAWAL

AI/ML Engineer | Machine Learning | Deep Learning | Python

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## SUMMARY

AI/ML Engineer with hands-on experience in machine learning, deep learning, and computer vision. Completed a one-year research project on AI-assisted design in nanophotonics, where I used neural networks and optimization algorithms to solve an inverse design problem. Skilled in Python, TensorFlow, and data handling. Looking for a full-time role to apply my skills in solving real-world problems using AI.

## EDUCATION

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <b>Master of Technology, Artificial Intelligence</b><br>Pandit Deendayal Energy University, Gandhinagar<br><i>Relevant coursework: Neural Networks, Pattern Recognition &amp; Machine learning, Essential Math in AI, Open CV, Time Series Data Analysis, AI in Finance</i> | June 2025<br>8.32 CGPA |
| <b>Bachelor of Engineering, Computer Engineering</b><br>Sal Engineering and Technical Institute, GTU                                                                                                                                                                        | July 2023<br>7.59 CGPA |

## TECHNICAL SKILLS

- **Languages/Tools:** Python, Jupyter Notebook, Git/GitHub
- **ML/DL Frameworks:** TensorFlow, Keras, Scikit-learn, XGBoost
- **Computer Vision:** OpenCV, Transfer Learning, InceptionV3
- **Data Handling:** Pandas, NumPy, Matplotlib, Seaborn
- **Optimization:** Differential Evolution, Model Evaluation (MSE,  $R^2$ , accuracy)

## PROFESSIONAL EXPERIENCE

- Maxgen Technologies Pvt. Ltd.:** Machine learning Intern Feb 2023 – Apr 2023
- Gained proficiency with Python libraries and tools such as TensorFlow, Keras, Scikit-learn, and Pandas.
  - *Internship Project: Skin Disease Detection*
  - Technology used: Python, TensorFlow, Keras, OpenCV, NumPy, Pandas, Jupyter Notebook, Google Colab.
- BrainyBeam Technologies Pvt. Ltd.:** Internship in Data Science and Machine learning Jun 2022 – Jul 2022
- *Internship Project: Recommendation system using Sentiment Analysis*
  - Technology used: Data preprocessing, NLP, Jupyter Notebook, Python, Kaggle

## ACADEMIC PROJECTS

### Fruit Classification (Fresh vs Rotten)

- Collected and preprocessed image dataset of fresh and rotten fruits
- Applied data augmentation techniques to increase training data variety
- Used InceptionV3 with transfer learning to build an image classifier
- Achieved 97% classification accuracy on validation data
- Technology Used: Python, TensorFlow, Keras, OpenCV, NumPy, Google Colab, Pandas

### House Price Prediction — Link

- Performed exploratory data analysis and cleaned raw housing data
- Engineered features and applied various regression models
- Used XGBoost for final model with optimized hyperparameters
- Achieved  $R^2$  score of 0.82 on the test dataset
- Technology Used: Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

## OTHER WORK EXPERIENCE

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### **Pandit Deendayal Energy University (Research and Teaching Assistant)**

- Assisted lab sessions for AI/ML subjects
- Helped students with coding and project guidance
- Supported professors with grading and project mentoring

## RESEARCH EXPERIENCE AND PUBLICATIONS

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### **AI-Assisted Design Discovery in Nanophotonics – PDEU — Link**

July 2024 – May 2025

- Solved an inverse design problem using Deep Neural Networks and Differential Evolution to optimize 8-layer DBR structures ( $\text{Sb}_2\text{S}_3/\text{SiO}_2$ )
- Replaced traditional simulation (15 minutes per run) with a DNN-based surrogate model that predicts spectral response in milliseconds.
- Reduced total optimization time from weeks to minutes, enabling faster and scalable nanophotonic device design.
- Output: 244-dimensional spectral response (reflectance/transmittance in amorphous and crystalline phases)

### **ETOT – I Conference, Andhra Pradesh — Link**

January 2025

- Presented our work as a poster at the conference

### **Upcoming Conference Submission**

Expected January 2026

- Manuscript already written and will be submitted soon

### **Journal Paper Preparation**

Ongoing

- Manuscript under preparation for journal submission

### **LeafDoc: Plant Leaf Disease Detection Using AI**

Jan 2025 – May 2025

- Worked on a plant leaf disease detection project using image classification techniques.
- Performed dataset preparation, preprocessing, and trained CNN-based models.
- Published the work in **IEEE Xplore** (AISTS 2025 Conference).
- DOI: 10.1109/AISTS66100.2025.11233219



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